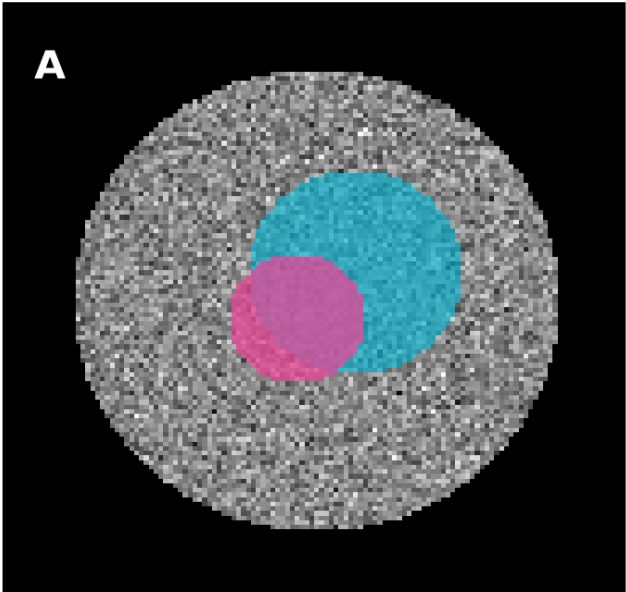
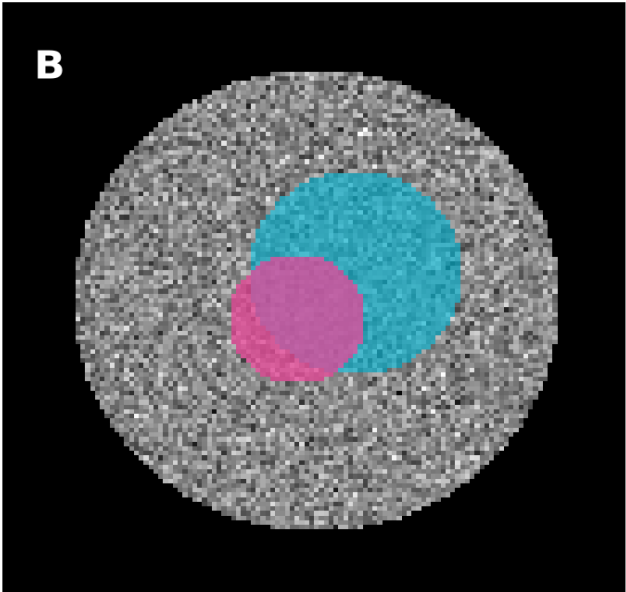


PhysioRANO NEURO-ONCOLOGY RADIOLOGY REPORT

Axial T1c (Post-Contrast)



Sagittal T2-FLAIR (Edema & Core)



CLINICAL HISTORY:

55-year-old Male presenting for post-radiotherapy Glioblastoma (GBM) evaluation at 60 days post-treatment.

- Molecular Profile: IDH1/2 **Mutated** | MGMT Promoter **Methylated**.
- Current Status: Routine follow-up scan monitoring, mild headaches.

FINDINGS:

BRAIN PARENCHYMA & 3D TUMOR SUB-REGIONS:

Multi-modal 3D MRI quantitative volumetric analysis demonstrates:

- **Whole Tumor (WT) Volume:** 24.50 cm³ (includes surrounding T2-FLAIR signal hyperintensity).
- **Tumor Core (TC) Volume:** 12.20 cm³.
- **Contrast-Enhancing Tumor (ET) Volume:** 18.50 cm³.

BIOPHYSICAL TUMOR GROWTH TRAJECTORY:

Biophysical neural network modeling evaluates the spatial invasion and proliferation dynamics:

- **Tissue Diffusion Speed:** Low spatial diffusion coefficient (0.1200 mm²/day).
- **Cell Proliferation Rate:** Modest cell proliferation rate (0.0400 day⁻¹).
- **Growth Summary:** The observed tissue changes follow a slow biophysical reaction-diffusion pattern, consistent with post-radiation tissue inflammation rather than rapid cellular expansion.

MULTIMODAL RECURRENCE & PSEUDOPROGRESSION ANALYSIS:

Integrated machine learning analysis evaluating clinical biomarkers, timeline post-treatment, and biophysical growth dynamics yields:

- **Probability of Pseudoprogression (Treatment Effect):** 78.4%
- Probability of True Tumor Progression: 14.2%
- Probability of Stable Disease: 7.4%
- **Key Contributing Factors:** IDH mutation status, favorable MGMT methylation, and timing within 12 weeks post-radiotherapy strongly favor treatment-induced tissue reaction.

RANO 2.0 CLINICAL RESPONSE EVALUATION:

Comparison with baseline MRI scans shows an enhancing volume shift from 12.00 cm³ to 18.50 cm³ (a +54.2% volumetric

change).

Under standard RANO 2.0 guidelines within the first 12 weeks post-radiotherapy, an enhancing expansion supported by high treatment-effect probability is classified as **Pseudoprogression (PsP)**.

3D CONDITIONAL GAN (cGAN) LONGITUDINAL FORECAST (+90 DAYS):

Generative biophysical 3D synthesis conditioned on spatial diffusion and proliferation predicts:

- **Projected Whole Tumor Volume: 384.17 cm³** (+86.7% trajectory shift).
- **Projected Enhancing Tumor Volume: 211.02 cm³** (+86.7% trajectory shift).
- **Clinical Trajectory Interpretation:** Projected morphological progression aligns with slow inflammatory stabilization under current adjuvant regimen.

IMPRESSION:

The enhancing volume shift (+54.2%) observed at 60 days post-radiotherapy in an IDH-mutated, MGMT-methylated glioblastoma patient is clinically consistent with **PSEUDOPROGRESSION (Radiation Necrosis / Treatment Effect)** rather than true disease recurrence.

CLINICAL RECOMMENDATION: Re-evaluate scan in 4-8 weeks.